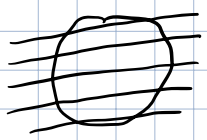
介质密度对注量影响.

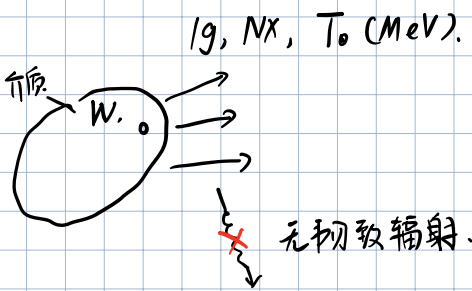
CEP, 初级辐射注量不变, 介质密度 ρ ↗ 1倍.

次级带电粒子注量? X2.



- 单位质量产生带电粒子数. X1
- 单位体积产生带电粒子数 X2
- 次级带电粒子射程.(cm) 1/2
- 次级带电粒子质量射程. X1
- 注量 X1
- X1

平衡注量



$$D_W \overset{CEP}{=} K_W = N T_0$$

$$= \int_0^{T_0} \Phi_T^e \left(\frac{dT}{\rho dx}\right)_W dT \quad \Rightarrow \quad \Phi_T^e = \frac{N}{\left(\frac{dT}{\rho dx}\right)_W}$$ 平衡注量.

$$\Phi^e = \int_0^{T_0} \frac{N}{[dT/(\rho dx)]_W} = N \mathcal{R}_{CSDA} = \rho N \frac{\mathcal{R}_{CSDA}}{\rho}.$$

$$\mathcal{R}_{CSDA} = \int_0^{T_0} \frac{dT}{-dT/\rho dx} \quad (g/cm^2)$$

$\Phi^e \propto N, \mathcal{R}_{CSDA}$

· 以质量厚度为单位，Z相差不大，阻止本领与射程相同.

介质密度变化，平衡注量不变.

进一步，非均匀源，而是均匀的X或γ外照射辐射场同样满足.

Fano theorem

$K_c = R_{in} - R_{out}^{nour}$

$D = (R_{in})_u - (R_{out})_u + (R_{in})_c$

$-(R_{out})_c$.

